Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science* Madison, WI
Awards: Dean's List (4 semesters) Expected May 2026
Coursework: Operating Systems, Algorithms, AI/ML, Statistics, Database Management, Data Structures **GPA: 3.7/4.0**

Experience

Mainframe Intern (Part-time) – *Wisconsin Department of Administration* **Jun 2025 – Present**

- Reduced manual triage time by 90% with a Python data pipeline that parsed system logs into structured datasets, flagged remediation status, and routed follow-ups across 5+ teams.
- Eliminated 4+ hours/week of manual reporting with a Python orchestrator using YAML-driven configuration, JSON Schema validation, Jinja2 templating, and automated email/SharePoint distribution.
- Deployed an IaC automation platform (Ansible, Terraform) adopted department-wide, streamlining 15+ batch job workflows across state mainframe systems.

Software Developer Intern (Part-time) – *Vijay Rekha Life Sciences* **Aug 2025 – Present**

- Shipped a production lab system (FastAPI, PostgreSQL) processing 300+ samples daily with automated validation, abnormality flagging, and Alembic-managed schema migrations.
- Accelerated clinical report validation by 85% using a batch processing pipeline (Python, pdfplumber, regex) that extracts and cross-validates patient data from 100+ PDF reports across multiple formats.

Computer Science Tutor – *Undergraduate Learning Center, UW–Madison* **Sep 2024 – Aug 2025**

- Mentored 600+ students across 8 CS courses and evaluated 160+ tutor candidates across 3 hiring cycles, designing technical interview questions for the Undergraduate Learning Center.

ERP Integration Intern – *United States Pharmacopeia* **Jun – Aug 2024**

- Improved data retrieval time by 70% with an Oracle ERP analytics tool (Python, SQL, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.
- Automated quarterly stakeholder reports, previously 1 week of manual work, with an ETL pipeline (Matplotlib, Seaborn) rendering visualizations into Word, PPTX, and PDF for executive distribution.

Projects

Office Hours Queue | *Python, Go/Gin, PostgreSQL, Redis, Docker* **Sep 2025 – Present**
Co-Founder · Advisor: Prof. Louis Oliphant (CS)

- Launched a distributed platform serving 500+ students across 6 containerized microservices with real-time SSE streaming, Redis pub/sub messaging, and PostgreSQL-backed state management.
- Designed a Redis pub/sub pipeline handling concurrent queue updates across 8 Gunicorn/gevent workers with keepalive heartbeats and ZSET-based tiered ordering synced to PostgreSQL for consistency.
- Migrated the Python/Flask backend to Go/Gin to eliminate GIL bottlenecks, leveraging goroutines and channels for concurrent SSE streaming and Redis pub/sub handling across real-time updates.
- Enabled one-command deployments for 6 services with Docker Compose and Traefik reverse proxy, automating CI/CD via GitHub Actions and Portainer webhooks with zero-downtime rollouts.

E-Beam 3D Printer Control System | *Python, Node.js, Express, GCP* **Jan – Aug 2025**
Morgridge Institute for Research (Prof. Kevin Eliceiri)

- Architected a concurrent, multi-threaded Python control system integrating 6+ hardware subsystems over Serial, Modbus RTU, and custom binary protocols with fault-tolerant exponential backoff and graceful degradation.
- Engineered a real-time monitoring dashboard (Node.js, Express, GCP) streaming sensor data with log2-based downsampling to visualize weeks of experiment data across 6 temperature sensors and 13 interlock points.
- Implemented a custom network protocol driver for the G9SP safety controller parsing 199-byte binary packets with bit-field decoding for 13 interlock channels, checksum validation, and exponential backoff on link failures.

Skills

Languages: Python, Go, Java, JavaScript, TypeScript, C, SQL, Bash
Infrastructure & DevOps: Docker, Docker Compose, Terraform, Ansible, Git, GitHub Actions, Traefik, Portainer, Gunicorn
Frameworks & Tools: Flask, FastAPI, Gin, Node.js/Express, React, Pytest
Databases & Cloud: PostgreSQL, Redis, MySQL, SQLite, Google Cloud Platform, Linux
Concepts: Distributed Systems, Reliability Engineering, Incident Response, Observability, Automation, Multi-threaded Programming, Fault-Tolerant Design, Network Protocol Design, CI/CD, REST APIs